

Does the text color impact reading speed?

PSTAT122: Design and Analysis of Experiments

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STUDENT NAME

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Due Date

Due Date: Monday, Dec 8, 2025, 11:59 PM

1 Introduction

The aim of the experiment is to find the effect of text color on reading speed.

Digital presentations—such as advertisements, websites, and social-media content—often rely on strategic color choices to emphasize key information and capture viewers’ attention. However, we are curious about whether different text colors influence how quickly people read and process information. Understanding whether certain colors slow down or speed up reading can inform better design decisions for digital interfaces, educational materials, and accessibility standards.

2 Experimental Design

Factors and Treatment Structure This experiment aims to study whether text color on white background influence reading speed . The main factor is Text Color in 4 levels(4 treatment):

1. YELLOW: #FFFF00
2. GREEN: #00FF00
3. RED: #CC0000

4. BLUE: #0000FF

Response Variable The response variable is Reading Time, measured in seconds. It is defined as the amount of time a participant needs to finish reading a similar difficult level text of approximately 250 words.

Fixed vs. Random Factors

Fixed Factor: Text Color Text color is a fixed factor because the four color levels were purposely chosen to represent color conditions. They are not randomly sampled from all possible colors, they are specific treatments that we want to compare in this experiment.

Random Factor: Participants

Participants are random factor. Four individuals from our friend group were invited to take part in the experiment. Since the analysis focuses on the effects of the text color rather than on these specific individuals, participants are modeled as a random effect. (Also known as blocking)

Design Type

A Randomized Complete Block Design (RCBD) is used in this experiment.

1. Randomize text colors (treatment) within each participant (block).

2. Each text color (treatment) appears exactly once within each block.

RCBD is appropriate because reading speed can vary greatly across individuals. Blocking on participants removes this variability and increases the ability to detect color effects.

Randomization

To reduce potential order effects, the order of the four text colors is randomized separately for each participant. This can be implemented by R function `sample()` to generate a random permutation of the colors:

```
1 colors <- c("Yellow", "Green", "Red", "Blue")
2
3 set.seed(122)
4 orders <- t(replicate(4, sample(colors)))
5
6 assignment <- as.data.frame(orders)
7 colnames(assignment) <- paste0("TextColor_", 1:4)
8 rownames(assignment) <- paste0("Participant_", 1:4)
9
10 assignment
```

	TextColor_1	TextColor_2	TextColor_3	TextColor_4
Participant_1	Blue	Yellow	Green	Red

Participant_2	Red	Blue	Yellow	Green
Participant_3	Green	Red	Yellow	Blue
Participant_4	Red	Yellow	Green	Blue

Replication

Each text color (treatment) appears once per participant(Blocking). Since there is 4 participant(Blocking), there is 4 replication for each treatment.

Blocking

Participant is blocking. Blocking controls for individual differences in baseline reading speed, attention level, and reading ability. By comparing treatment effects within the same participant rather than across different individuals, blocking substantially reduces variability caused by personal differences and allows more precise estimation of the treatment effect.

Sample Size 4 treatment levels(text color)

4 blocks (participants)

1 observation per treatment per block

Total = 16 observations

Important: Since we only want to analyze the different between text color on reading speed, we find 4 similar texts and randomly assign color to each of them. (One specific text for each color) Those texts have similar topics, length and reading difficulty.

```

1 colors <- c("Red", "Blue", "Yellow", "Green")
2
3 set.seed(122)
4 color_assign <- sample(colors)
5
6 names(color_assign) <- c("Text_A", "Text_B", "Text_C", "Text_D")
7
8 color_assign

```

Text_A	Text_B	Text_C	Text_D
"Green"	"Red"	"Blue"	"Yellow"

```

1 assignment

```

	TextColor_1	TextColor_2	TextColor_3	TextColor_4
Participant_1	Blue	Yellow	Green	Red

Participant_2	Red	Blue	Yellow	Green
Participant_3	Green	Red	Yellow	Blue
Participant_4	Red	Yellow	Green	Blue

3 Data Collection

Location: The study was conducted in Siyuan’s living room.

Date and Time: The experiment took place on December 6, 2025, at 5:02 PM.

Recorder: Data were recorded by Siyuan Lyu

Participants: 4 UCSB students with Chinese as first language, 3 male 1 female

Settings: Participants read the texts on a 14-inch MacBook Pro using the default display settings with screen brightness set to 100%. All texts were prepared in Google Docs, formatted in 12-point Arial font and presented in a double-sided layout and page zoom is 100%.

Procedure: Before each trial, the participant was instructed to close their eyes. The timer was started when the participant opened their eyes and began reading the text, and was stopped when the participant verbally indicated that they had finished reading.

The same procedure was then repeated for the second text, beginning again with eyes closed.

After a participant completed all assigned texts, the reading time for each text was recorded. The procedure was then repeated for the next participant.

Adjustment: Initially, all timers were started and stopped by the recorder. However, we observed that after participants opened their eyes, there was a brief delay between the moment of eye opening and the moment they actually began reading. Additionally, at the end of each reading session, the recorder’s reaction time in stopping the timer introduced nontrivial measurement error.

To minimize this source of timing inaccuracy, participants were later instructed to start and stop the timer themselves.

Furthermore, due to variations of lighting within the room, all participants were asked to read at the same table and seated in the same chair to control for potential differences in brightness that could influence reading performance.

Data are stored in the dataframe below (seconds):

```

1 n_participants <- 4
2 participants <- paste0("P", 1:n_participants)
3
4
5 experiment_time <- data.frame(
6   Participant = participants,
7   Time_A_Green = c(29.90,30.04,24.61,22.66),
8   Time_B_Red = c(23.90,26.86,30.01,21.33),
9   Time_C_Blue = c(25.85,32.30,23.88,20.66),
10  Time_D_Yellow = c(42.66,40.88,41.83,26.75)
11 )
12
13 experiment_time

```

	Participant	Time_A_Green	Time_B_Red	Time_C_Blue	Time_D_Yellow
1	P1	29.90	23.90	25.85	42.66
2	P2	30.04	26.86	32.30	40.88
3	P3	24.61	30.01	23.88	41.83
4	P4	22.66	21.33	20.66	26.75

Visualization:

```

1 long_df <- experiment_time %>%
2   pivot_longer(
3     cols = starts_with("Time_"),
4     names_to = c("Paragraph", "Color"),
5     names_pattern = "Time_{[A-D]}_(.*)",
6     values_to = "Time"
7   )
8
9 summary_table <- long_df %>%
10  group_by(Color) %>%
11  summarise(
12    Mean_Time = mean(Time),
13    SD_Time = sd(Time),
14    .groups = "drop"
15  )
16
17 ggplot(summary_table, aes(x = Color, y = Mean_Time, fill = Color)) +
18   geom_col(width = 0.6) +
19   geom_errorbar(

```

```

20   aes(ymin = Mean_Time - SD_Time,
21       ymax = Mean_Time + SD_Time),
22   width = 0.2,
23   linewidth = 0.8
24 ) +
25 scale_fill_manual(values = c(
26   "Blue"    = "blue",
27   "Green"   = "green3",
28   "Red"     = "red",
29   "Yellow"  = "yellow"
30 )) +
31 labs(
32   title = "Figure 1:Mean Reading Time by Text Color",
33   x = "Color",
34   y = "Mean Reading Time (seconds)"
35 ) +
36 theme_minimal() +
37 theme(text = element_text(size = 14))

```

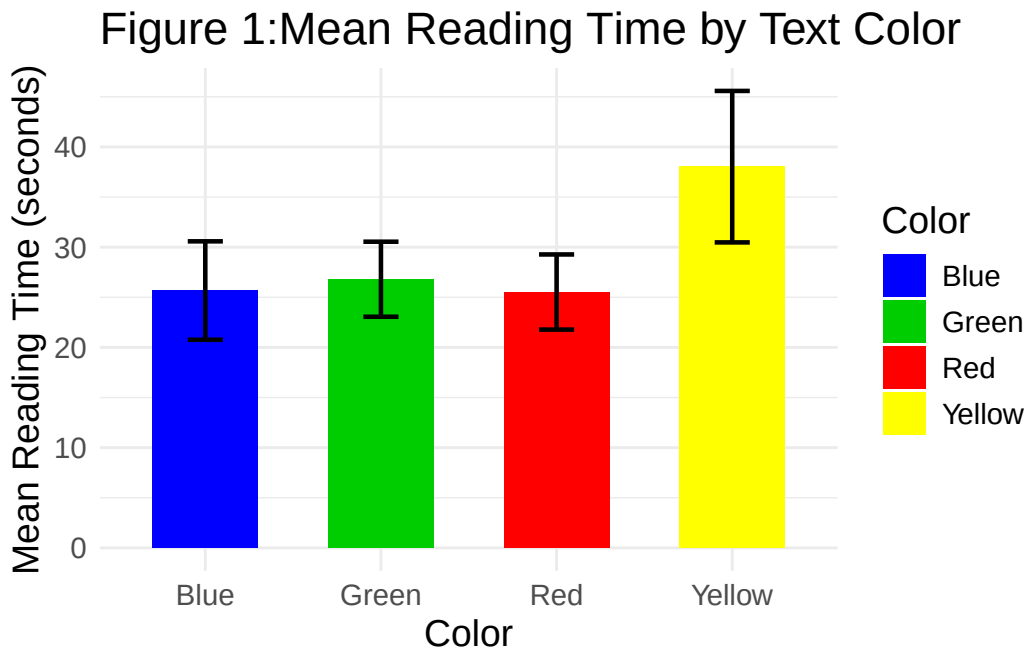


Figure 1 summarizes the mean reading times and standard deviations for each text color condition. Reading performance was relatively consistent across the Blue, Green, and Red text conditions, with mean times of 25.67 seconds, 26.80 seconds, and 25.52 seconds, respectively. These three colors also showed comparable variability, with standard deviations ranging from approximately 3.74 to 4.91 seconds.

In contrast, the Yellow text condition produced substantially longer reading times, with a mean of 38.03 seconds, which is roughly 11–13 seconds slower than the other colors. The Yellow condition also showed the greatest variability (SD = 7.56 seconds), indicating more inconsistent performance across participants.

4 Analysis

To begin the analysis, the data were first reshaped into a long format so that each reading-time measurement corresponded to a unique combination of Participant (block) and Text Color (treatment). This structure allows us to easily summarize, visualize, and model the data using standard tools for randomized complete block designs.

As an initial exploratory step, we calculated the mean and standard deviation of reading time for each color condition. These basic descriptive statistics provide a first look at how performance varies across treatments. The summary revealed that Blue, Red, and Green all had relatively similar mean reading times (approximately 25–27 seconds) with moderate variability, while Yellow had a substantially higher mean (around 38 seconds) and noticeably larger standard deviation. This suggests that participants were not only slower when reading Yellow text, but also less consistent across trials.

To further visualize these patterns, a grouped mean plot with error bars (Figure 1) was produced. This clearly shows the separation between Yellow and the other three colors. A complementary boxplot also highlights the distributional differences: Blue, Red, and Green exhibit relatively tight, overlapping distributions, whereas Yellow displays a higher median, wider spread, and several higher values. Taken together, the descriptive statistics and visualizations indicate that Yellow is an outlier in terms of reading speed, while the other colors behave similarly.

Because this experiment follows a Randomized Complete Block Design (RCBD) with Participants as blocks and Text Color as the treatment factor, a two-way ANOVA model without interaction was used

$$\text{Time}_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij},$$

where

- τ_i is the fixed effect of the i -th Text Color level,
- β_j is the random blocking effect for participant j ,
- ϵ_{ij} represents random error.

The null hypothesis tested is:

$$H_0 : \tau_{\text{Red}} = \tau_{\text{Blue}} = \tau_{\text{Yellow}} = \tau_{\text{Green}} \quad (\text{no difference in mean reading time across colors})$$

versus

$$H_a : \text{At least one color differs.}$$

4.0.1 ANOVA Results

```
1 aov_rcbd <- aov(Time ~ Color + Participant, data = long_df)
2 summary(aov_rcbd)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Color	3	438.1	146.02	11.711	0.00184 **
Participant	3	215.5	71.83	5.761	0.01765 *
Residuals	9	112.2	12.47		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The ANOVA indicates that:

Text Color has a statistically significant effect on reading time ($F = 11.711$, $p = 0.00184$).

Participant also has a significant blocking effect ($F = 5.761$, $p = 0.01765$), confirming that individuals differ in baseline reading speed and justifying the RCBD structure.

4.0.2 Post-Hoc Comparisons

To determine which colors differ significantly, Tukey's HSD test was conducted:

```
1 TukeyHSD(aov_rcbd, "Color")
```

Tukey multiple comparisons of means
95% family-wise confidence level

Fit: aov(formula = Time ~ Color + Participant, data = long_df)

\$Color		diff	lwr	upr	p adj
Green-Blue	1.1300	-6.664733	8.924733	0.9674522	
Red-Blue	-0.1475	-7.942233	7.647233	0.9999201	
Yellow-Blue	12.3575	4.562767	20.152233	0.0036019	
Red-Green	-1.2775	-9.072233	6.517233	0.9542303	
Yellow-Green	11.2275	3.432767	19.022233	0.0066897	
Yellow-Red	12.5050	4.710267	20.299733	0.0033286	

The Tukey HSD post-hoc comparisons provide insight into which specific color pairs differ significantly in mean reading time. The comparisons among Blue, Red, and Green all have large p-values (Green-Blue: $p = 0.967$, Red-Blue: $p = 0.999$, Red-Green: $p = 0.954$), indicating that these three colors do not differ meaningfully from one another. Their confidence intervals also span zero, suggesting no detectable difference in reading speed between any pair within this group.

In contrast, all comparisons involving Yellow show statistically significant differences. Yellow-Blue (diff = 12.36 sec, $p = 0.0036$), Yellow-Green (diff = 11.23 sec, $p = 0.0067$), and Yellow-Red (diff = 12.51 sec, $p = 0.0033$) all have positive differences with confidence intervals entirely above zero. This indicates that participants consistently took around 11–13 seconds longer to read Yellow text compared to any of the other three colors.

Overall, the Tukey results confirm that Yellow is the only color that significantly slows down reading, while Blue, Red, and Green do not differ significantly from each other. This reinforces the conclusion that the observed color effect found in the ANOVA is driven entirely by the Yellow condition.

5 Conclusions

The results of this experiment provide clear and consistent evidence that **text color affects reading speed**, and that this effect is driven almost entirely by the **Yellow** text condition. The descriptive statistics, visualizations, ANOVA results, and Tukey post-hoc comparisons all support the same conclusion: participants read significantly more slowly when the text was yellow, while the other three colors—Blue, Red, and Green—produced similar reading speeds.

Across all participants, Yellow text increased reading time by approximately **11–13 seconds** relative to the other colors. This corresponds to a substantial **40–50% slowdown**, highlighting a meaningful and practically important effect. In contrast, the pairwise comparisons among Blue, Red, and Green showed no statistically significant differences. Their confidence intervals overlapped widely, and all p-values were near 1.0, indicating that these colors do not materially differ from each other in terms of reading performance under the conditions of this experiment.

The Randomized Complete Block Design (RCBD) successfully controlled for individual differences in baseline reading speed, as confirmed by the significant participant (block) effect in the ANOVA. This blocking structure allowed us to isolate the impact of the color factor more precisely and strengthened the evidence that the color effect is real and not simply due to individual variability.

5.1 Limitations

While the results are strong, several limitations should be acknowledged:

- **Small sample size ($n = 4$ participants):** This limits the external validity and makes it difficult to generalize the findings to a broader population.
- **Homogeneous participant group:** All participants had similar backgrounds and reading abilities, which may reduce variability and limit generalizability.
- **Text differences:** Although the four texts were designed to be similar, subtle variations in content may still have influenced reading time.
- **Environmental constraints:** Lighting and screen conditions were manually controlled but could still vary slightly across trials.
- **Limited color range:** Only four specific colors were tested; variations in brightness, contrast, or alternative hues were not explored.

5.2 Possible Improvements

Future studies could enhance reliability and depth by:

- Increasing the number of participants to improve statistical power and generalizability.
- Using **identical text passages** for all conditions with color counterbalancing to ensure that content does not influence reading time.
- More systematically standardizing environmental conditions (lighting, monitor calibration, seating distance).
- Testing additional colors or varying contrast levels (e.g., darker vs. lighter yellow, different shades of red or blue).
- Including **comprehension checks** or **eye-tracking measures** to assess cognitive load, not just reading speed.

Overall, the experiment provides strong evidence that **bright yellow text significantly slows reading**, while red, blue, and green have no detectable impact on reading speed. These findings can inform design choices in digital interfaces, educational materials, and visual presentation standards.